ACT SUBJECT TEST & DIGITAL SAT MATH COURSES

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ACT2 • ACT SUBJECT TEST MATH LEVEL 1 • PRACTICE EXAM

ACT2 Exam 3

Full Practice Test • Test #3

Chords, prisms, inequalities, algebra & functions. Work through all 50 questions under timed conditions, then check your answers with the answer key and the step-by-step solutions at the end.

50	MCQ	60 min
QUESTIONS	MULTIPLE CHOICE	TIME LIMIT

NAME _____ DATE _____ SCORE _____ /50

How to get the most from this practice test

- **Read** every question slowly and underline the key words.
- **Draw** a quick sketch for every geometry question before you solve it.
- **Solve** step by step and show your work.
- **Check** your answer by substituting it back before you choose.

RF

YOUR MATH COACH

Eng. Rabaa Farrag

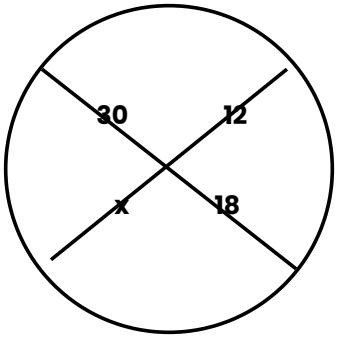
ACT, Digital SAT & EST Math expert with **23 years of experience**, helping students master math and reach **700+**.

ACT SUBJECT TEST • MATH LEVEL 1 • TEST #3

Practice Questions

50 QUESTIONS

1 Find the value of x in the circle shown below:



A) 54
B) 540
C) 360
D) 45

2 Factorize $4x^3y^2(2y^4x + 6x^3y^7)$:

A) $8x^3y^8 + 24x^9y^{14}$
B) $8x^4y^6 + 24x^6y^9$
C) $6x^4y^6 + 10x^6y^9$
D) $6x^3y^8 + 10x^9y^{14}$

3 What is the minimum number of points required to uniquely define a plane?

A) 4
B) 3
C) 2
D) 1

4 If $-3x + 7 < 16$, which of the following must be true?

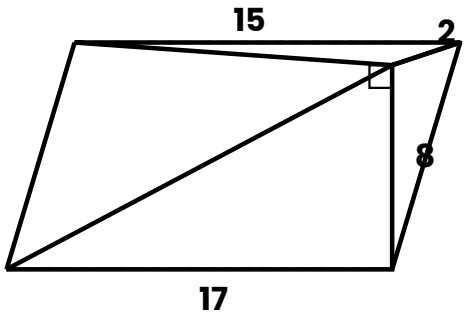
A) $x > -3$
B) $x < -3$
C) $x < -9$
D) $x > -9$

5 What are the solution(s) to the equation $5|x - 2| + 1 = 16$?

A) $x = 3$ only
B) $x = 5$ only
C) $x = 5$ and $x = 3$
D) $x = 5$ and $x = -1$

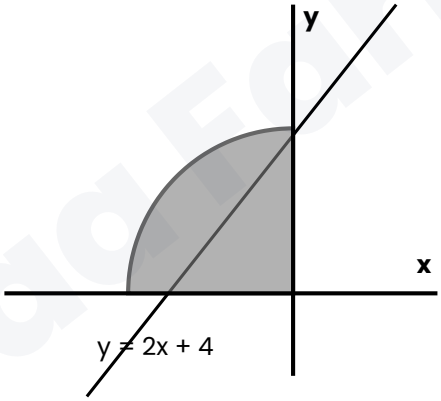
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6 Find the volume of the right triangular prism shown below:



(A) 100
(B) 120
(C) 240
(D) 480

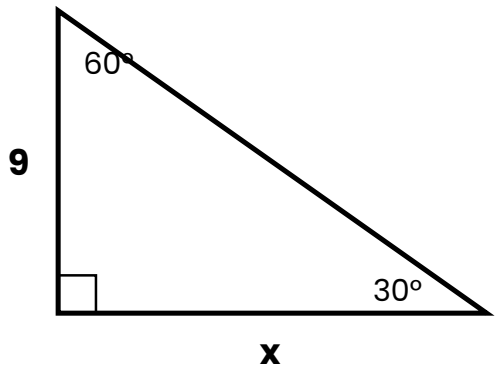
7 Find the perimeter of the figure below, bounded by the coordinate axes and the line $y = 2x + 4$ forming a quarter circle sector:



(A) $\pi\sqrt{5}$
(B) 6
(C) $6 + \pi\sqrt{5}$
(D) $\frac{5}{2}\pi + 4$

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8 In the $30° - 60° - 90°$ right triangle shown below, find the value of x:



A $9\sqrt{3}$
B 4.5
C 18
D 9

9 A rectangle has a length that is 6 units more than double its width. If the area of this rectangle is 80, find the length of the rectangle.

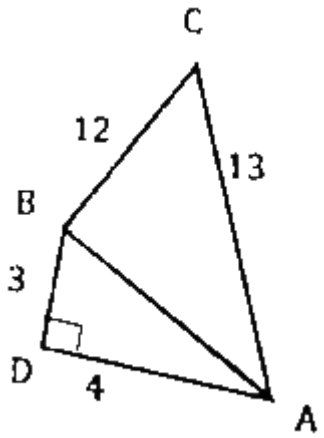
A 8
B 16
C 40
D 80

10 You have 5 strawberry donuts, 6 chocolate donuts, 3 glazed donuts, and 4 caramel donuts. If you choose a donut randomly, what is the probability that it is strawberry?

A $\frac{2}{9}$
B $\frac{1}{6}$
C $\frac{1}{3}$
D $\frac{5}{18}$

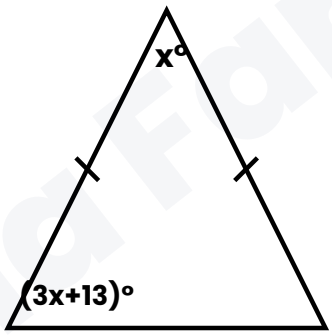
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11 What kind of triangle is triangle ABC shown in the figure below?



(A) Acute triangle
(B) Obtuse triangle
(C) Right triangle
(D) Isosceles triangle

12 The triangle below is an isosceles triangle with equal legs. Find the value of x:



(A) $x = 154$
(B) $x = 7$
(C) $x = 22$
(D) $x = 34$

13 A initial population of 6 rabbits doubles every year. How many rabbits will there be in the population after 10 years?

(A) 12
(B) 1200
(C) 6144
(D) 1024

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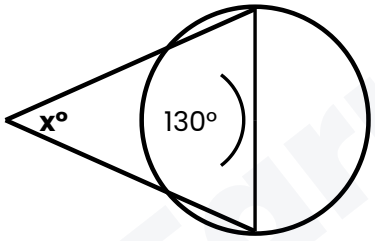
14 Solve the compound inequality: $2x - 8 \leq x - 4 < 3(x + 6)$

(A) $-11 < x < 4$
(B) $-11 > x > 4$
(C) $-11 < x \leq 4$
(D) $-11 \leq x < 4$

15 If the volume of a cube is equal to 2 times its total surface area, what is the side length s of the cube?

(A) $s = -12$
(B) $s = 12$
(C) $s = -6$
(D) $s = 6$

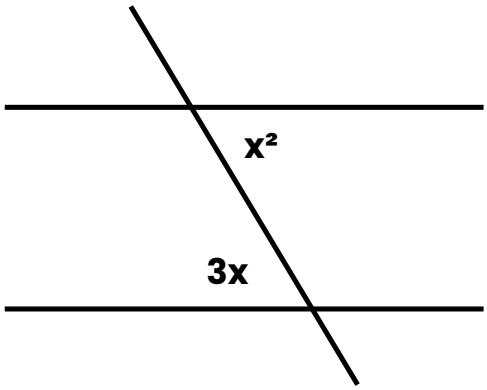
16 In the figure below, two tangents are drawn to a circle from an external point forming an angle $x°$. If the central angle subtended by the points of tangency is $130°$, find the value of x:



(A) $x = 20°$
(B) $x = 50°$
(C) $x = 90°$
(D) $x = 130°$

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17 Two parallel lines are intersected by a transversal, forming consecutive interior angles measuring x^2 degrees and $3x$ degrees. Solve for x:

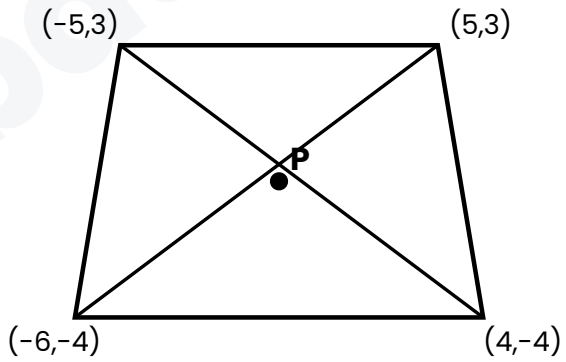


A) 45
B) 180
C) 15
D) 12

18 Solve for x: $2(x + 4) = 10$

A) 3
B) -1
C) 1
D) 2

19 Find the coordinates of point P where the two diagonals of the rectangle with vertices $(-5, 3)$, $(5, 3)$, $(-6, -4)$, and $(4, -4)$ intersect:



A) $(-1, -1)$
B) $(1, 1)$
C) $(-\frac{1}{2}, -\frac{1}{2})$
D) $(\frac{1}{2}, \frac{1}{2})$

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20 Which line segment always divides any triangle into two triangles of equal area?

A. A median
B. An altitude
C. An angle bisector
D. A perpendicular bisector of a side

21 In a college with 14,000 students, 490 students are majoring in mathematics. What percentage of the student body do the mathematics majors represent?

A. 0.0035%
B. 0.035%
C. 0.35%
D. 3.5%

22 Given that $\cos^2 x = \frac{5}{12}$, what is the value of $\sin^2 x$?

A. $\frac{5}{13}$
B. $\frac{7}{12}$
C. $\frac{5}{7}$
D. $\frac{12}{5}$

23 Michael constructed two parallel lines intersected by a transversal. Which of the following statements is a valid conjecture Michael can make about the resulting angles?

A. Pairs of same-side interior angles are supplementary.
B. Pairs of alternate interior angles are supplementary.
C. Pairs of alternate exterior angles are supplementary.
D. Pairs of corresponding angles are supplementary.

24 The endpoints of directed line segment PQ are $P(-7, -5)$ and $Q(5, 3)$. What are the coordinates of point A on PQ that divides PQ in the ratio $1 : 3$?

A. $A(-1, 1)$
B. $A(2, 1)$
C. $A(3, 2)$
D. $A(-4, -3)$

25 Zahra spent $20.50 on 10 party favors for her party. Puzzle books cost $1.75 each for boys, and magic tricks cost $2.25 each for girls. How many boys and girls attended the party?

A. 4 boys and 6 girls
B. 5 boys and 5 girls
C. 6 boys and 4 girls
D. 7 boys and 3 girls

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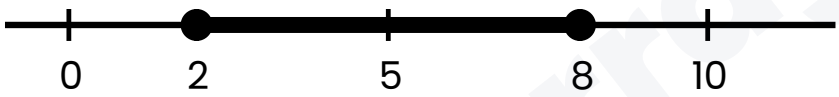
26 John is comparing two sales job offers. Company ABC offers a base salary of $45,000. Company XYZ offers $38,000 plus a 2% commission on sales. For what total sales amount (X) is the salary at XYZ greater than ABC?

(A) $X > 7,000$
(B) $X > 35,000$
(C) $X > 70,000$
(D) $X > 350,000$

27 Write an equation for the line with y-intercept 2 that is perpendicular to the line $3x + y = 6$:

(A) $y = -3x + 2$
(B) $y = -3x - 2$
(C) $y = \frac{1}{3}x + 2$
(D) $y = \frac{1}{3}x - 2$

28 Which compound inequality is represented by the shaded line segment between 2 and 8 on the number line below?



(A) $x \geq 2$ and $x \leq 8$
(B) $x \geq 2$ or $x \geq 8$
(C) $x \geq 2$ or $x \leq 8$
(D) $x \leq 2$ or $x \leq 8$

29 Big Sky Mountain charges $5 per hour plus $50 insurance. Powder Hills charges $10 per hour plus $30 insurance. For how many hours of lessons is the total cost identical at both resorts?

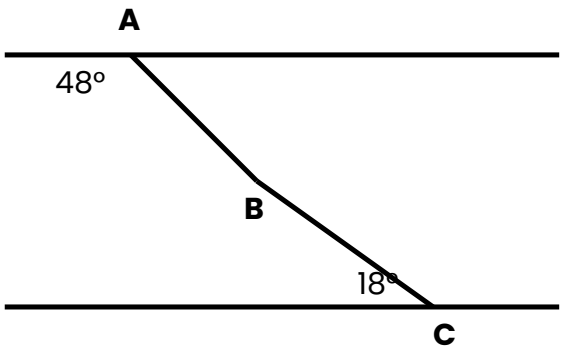
(A) 3 hours
(B) 4 hours
(C) 5 hours
(D) 6 hours

30 Find the intersection of set $W = \{0, 1, 2, 3, \dots\}$ (whole numbers) and set $P = \{\dots, -2, -1, 1, 2, 3, \dots\}$ (nonzero integers):

(A) $\{\}$
(B) $\{0\}$
(C) The set of negative integers
(D) The set of positive integers

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31 In the figure below, line FAD is parallel to line EHC. If $m\angle FAB = 48°$ and $m\angle ECB = 18°$, find $m\angle ABC$:



(A) $18°$
(B) $48°$
(C) $66°$
(D) $114°$

32 A carton holds X eggs. Employees have E eggs to pack. Using the formula $\frac{E}{X} = C + \frac{3}{X}$, solve for C, the number of full cartons packed:

(A) $C = \frac{E-3}{X}$
(B) $C = \frac{E}{X} - 3$
(C) $C = \frac{E}{X-3}$
(D) $C = 3 - \frac{E}{X}$

33 In the identity $4x^2 + ax = 2x(2x + 1)$, what is the value of a?

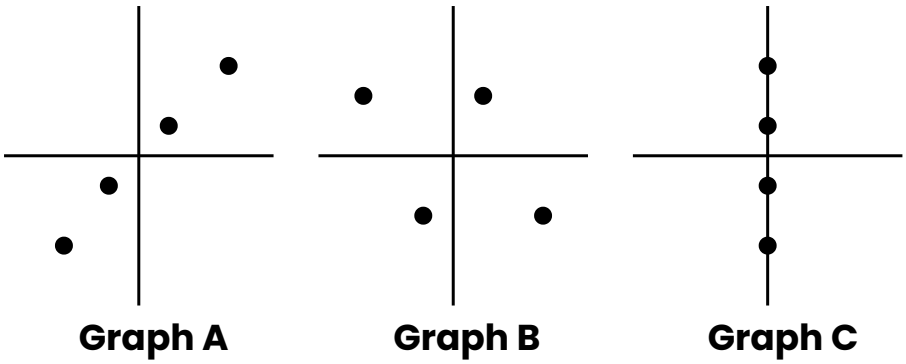
(A) 2
(B) 4
(C) 6
(D) 8

34 If $\angle A$ is an acute angle, how many degrees larger is its supplement than its complement?

(A) $-90°$
(B) $0°$
(C) $90°$
(D) $180°$

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35 Determine which of the scatter plot graphs below represent a function:



(A) None of the graphs are functions.
(B) All of the graphs are functions.
(C) Graphs A and B are functions.
(D) Graphs B and C are functions.

36 If $0.002 \leq x \leq 0.2$ and $5 \leq y \leq 25$, what is the maximum possible value of $\frac{x}{y}$?

(A) 0.04
(B) 0.4
(C) 4
(D) 40

37 Write an equation in slope-intercept form for the line passing through $(3, 7)$ and $(7, 4)$:

(A) $y = -\frac{3}{4}x + \frac{37}{4}$
(B) $y = \frac{3}{4}x + \frac{37}{4}$
(C) $y = -\frac{4}{3}x + \frac{37}{4}$
(D) $y = -\frac{3}{4}x + \frac{4}{37}$

38 If true statements have truth value 1 and false statements have truth value 0, what is the truth value of the compound statement: "All squares are rectangles, OR all squares are circles"?

(A) -1
(B) 0
(C) $\frac{1}{2}$
(D) 1

39 In any right triangle, the altitude drawn to the hypotenuse always:

(A) lies inside the triangle
(B) lies outside the triangle
(C) bisects the hypotenuse
(D) bisects the right angle

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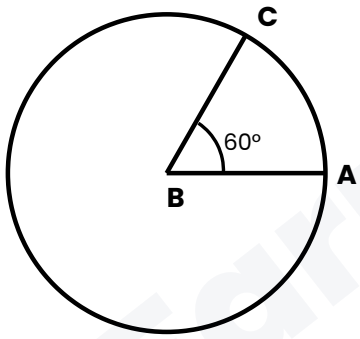
40 What is the sum of the measures of the interior angles of a 14-sided polygon?

(A) 1,980
(B) 2,160
(C) 2,520
(D) 2,880

41 Simplify the expression $\frac{a^{15}b^{12}-a^5b^8}{a^3b^2}$ for $a \neq 0, b \neq 0$:

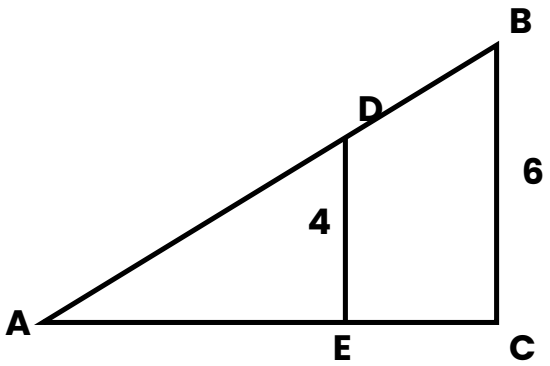
(A) $a^5b^6 - a^5b^4$
(B) $a^{12}b^{10} - a^2b^6$
(C) a^6b^4
(D) a^7b^2

42 If minor arc $\overset{\frown}{AC} = 4\pi$ cm in a circle with central angle $60°$, find the area in square centimeters of sector ABC:



(A) 4π
(B) 12π
(C) 24π
(D) 40π

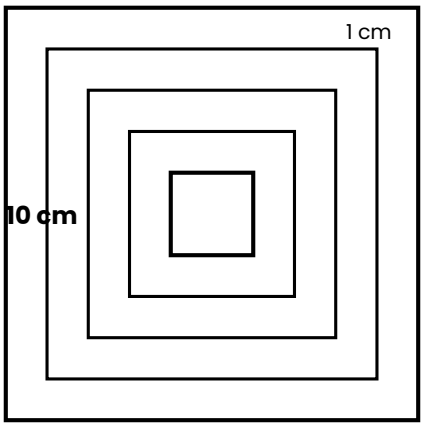
43 In right triangle ABC with $DE \parallel BC$, $DE = 4$, $BC = 6$, and $AC = 9$. What is the measurement of EC?



(A) 3
(B) 5
(C) 6
(D) $\frac{27}{2}$

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44 In the figure, 5 nested squares are positioned such that the distance between adjacent squares is 1 cm. If the side length of the largest square is 10 cm, what is the ratio of the area of the largest square to that of the smallest square?



A) 2:1
B) 5:3
C) 25:9
D) 25:1

45 Which expression is equivalent to $(h^2 + 9h - 1)(-4h + 3)$?

A) $-4h^3 - 33h^2 + 31h - 3$
B) $4h^3 + 39h^2 - 23h - 3$
C) $-4h^3 - 39h^2 + 23h + 3$
D) $4h^3 + 33h^2 - 31h + 3$

46 What expression is equivalent to $x^2 - 17x - 60$?

A) $(x - 20)(x + 3)$
B) $(x - 5)(x - 12)$
C) $(x - 20)(x - 3)$
D) $(x + 5)(x - 12)$

47 The formula $A = p + prt$ models the balance in a savings account. Which equation is correctly solved for interest rate r?

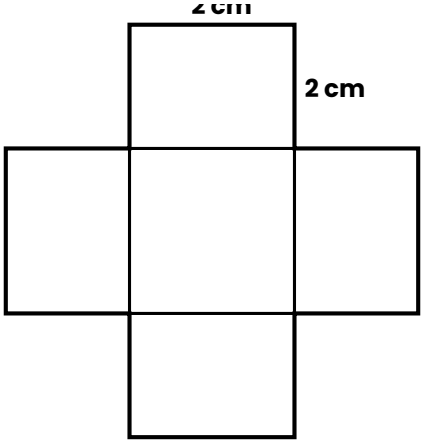
A) $r = Apt$
B) $r = A - 2pt$
C) $r = \frac{A}{2pt}$
D) $r = \frac{A-p}{pt}$

48 Which expression is equivalent to $\frac{(q^4)^{-3}}{q^{-15}}$ for all defined values of q?

A) q^{27}
B) $\frac{1}{q^{27}}$
C) q^3
D) $\frac{1}{q^3}$

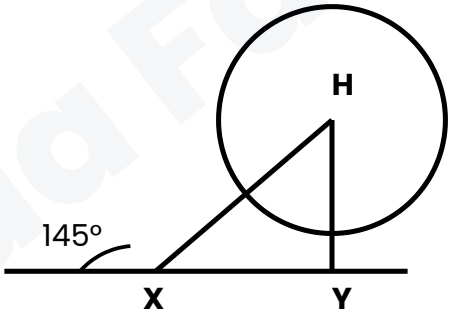
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49 The figure below is a cross shape composed of 5 identical squares of side length 2 cm. What is the total area of the figure?



(A) 8 cm^2
(B) 16 cm^2
(C) 20 cm^2
(D) 24 cm^2

50 In the circle with center H, line $\overline{XY}$ is tangent to the circle at point Y. If the exterior angle at X measures $145°$, find $m\angle XHY$:



(A) $25°$
(B) $35°$
(C) $90°$
(D) $55°$

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TEST #3 • CHECK YOUR WORK

Answer Key

50 ANSWERS

Q	ANSWER	Q	ANSWER
1	D	26	D
2	B	27	C
3	B	28	A
4	A	29	B
5	D	30	D
6	B	31	C
7	C	32	A
8	A	33	A
9	B	34	C
10	D	35	C
11	C	36	A
12	C	37	A
13	C	38	D
14	C	39	A
15	B	40	B
16	B	41	B
17	D	42	C
18	C	43	A
19	C	44	D
20	A	45	A
21	D	46	A
22	B	47	D
23	A	48	C
24	D	49	C
25	A	50	D

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TEST #3 • STEP-BY-STEP

Detailed Solutions

50 SOLUTIONS

1 **Answer: D** 45

By the Intersecting Chords Theorem, the products of the segments of intersecting chords are equal:

$$30 \cdot 18 = 12 \cdot x$$

$$540 = 12x \implies x = 45$$

2 **Answer: B** $8x^4y^6 + 24x^6y^9$

Expand the term by distributing $4x^3y^2$ across the terms inside the parentheses:

$$4x^3y^2 \cdot 2y^4x + 4x^3y^2 \cdot 6x^3y^7 = 8x^4y^6 + 24x^6y^9$$

3 **Answer: B** 3

Step 1: A plane is fixed by points that are not all on one line.
Step 2: Two points only fix a line (infinitely many planes contain it).
Step 3: Three non-collinear points determine exactly one plane — the minimum is 3.

4 **Answer: A** $x > -3$

Subtract 7 from both sides:

$$-3x < 9$$

Divide by -3 (and flip the inequality sign):

$$x > -3$$

5 **Answer: D** $x = 5$ and $x = -1$

Isolate the absolute value term:

$$5|x - 2| = 15 \implies |x - 2| = 3$$

Set up the two linear equations:
1) $x - 2 = 3 \implies x = 5$
2) $x - 2 = -3 \implies x = -1$

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6 **Answer: B** 120

The base of the prism is a right triangle with hypotenuse 17 and vertical leg 8. The missing horizontal leg b is:

$$b = \sqrt{17^2 - 8^2} = \sqrt{289 - 64} = \sqrt{225} = 15$$

Area of the triangular base:

$$\text{Area} = \frac{1}{2} \times 15 \times 8 = 60$$

Volume of the prism:

$$\text{Volume} = \text{Base Area} \times \text{depth} = 60 \times 2 = 120$$

7 **Answer: C** $6 + \pi\sqrt{5}$

The line $y = 2x + 4$ has y-intercept at $(0, 4)$ and x-intercept at $(-2, 0)$. The straight sides along the axes measure 2 units and 4 units (total straight perimeter $= 2 + 4 = 6$).
The radius of the circular arc is $r = \sqrt{(-2)^2 + 4^2} = \sqrt{20} = 2\sqrt{5}$.
The arc length of the quarter circle is:

$$\text{Arc} = \frac{1}{4}(2\pi r) = \frac{\pi(2\sqrt{5})}{2} = \pi\sqrt{5}$$

Total perimeter $= 6 + \pi\sqrt{5}$.

8 **Answer: A** $9\sqrt{3}$

In a $30° - 60° - 90°$ right triangle, the side opposite the $30°$ angle is $a = 9$. The side opposite the $60°$ angle is:

$$x = a\sqrt{3} = 9\sqrt{3}$$

9 **Answer: B** 16

Let w be the width. The length is $l = 2w + 6$.

$$\text{Area} = w(2w + 6) = 80$$

$$2w^2 + 6w - 80 = 0 \implies w^2 + 3w - 40 = 0$$

$$(w + 8)(w - 5) = 0 \implies w = 5$$

Substitute $w = 5$ to find the length:

$$l = 2(5) + 6 = 16$$

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10 **Answer: D** $\frac{5}{18}$

Total number of donuts $= 5 + 6 + 3 + 4 = 18$.
Number of strawberry donuts $= 5$.

$$P(\text{strawberry}) = \frac{5}{18}$$

11 **Answer: C** Right triangle

Step 1: From the figure, $\triangle BDA$ has a right angle at D with $BD = 3$ and $DA = 4$, so $BA = \sqrt{3^2 + 4^2} = 5$.
Step 2: Now $\triangle ABC$ has sides $BA = 5$, $BC = 12$, $CA = 13$.
Step 3: $5^2 + 12^2 = 25 + 144 = 169 = 13^2$, so by the converse of the Pythagorean theorem $\angle ABC = 90°$: a right triangle.

12 **Answer: C** $x = 22$

Since the two legs are marked as equal, the two base angles are also equal, each measuring $(3x + 13)°$.
The sum of interior angles in a triangle is $180°$:

$$x + 2(3x + 13) = 180$$

$$x + 6x + 26 = 180 \implies 7x = 154 \implies x = 22$$

13 **Answer: C** 6144

Exponential growth model for doubling every year:

$$N = N_0 \cdot 2^t = 6 \cdot 2^{10} = 6 \cdot 1024 = 6144$$

14 **Answer: C** $-11 < x \leq 4$

Solve the two parts separately:
1) $2x - 8 \leq x - 4 \implies x \leq 4$
2) $x - 4 < 3x + 18 \implies -22 < 2x \implies x > -11$
Combining both gives: $-11 < x \leq 4$.

15 **Answer: B** $s = 12$

Volume of cube $= s^3$. Total surface area of cube $= 6s^2$.

$$s^3 = 2(6s^2) = 12s^2$$

Dividing by s^2 (since $s > 0$):

$$s = 12$$

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16 **Answer: B** $x = 50^\circ$

The quadrilateral formed by the center, the two points of tangency, and the external point has two right angles (90°) at the points of tangency. Therefore, the central angle and the external angle between the tangents are supplementary:

$$x + 130^\circ = 180^\circ \implies x = 50^\circ$$

17 **Answer: D** 12

Consecutive interior angles formed by parallel lines cut by a transversal are supplementary (sum to 180°):

$$x^2 + 3x = 180 \implies x^2 + 3x - 180 = 0$$

$$(x + 15)(x - 12) = 0$$

Since an angle measure must be positive, $x = 12$.

18 **Answer: C** 1

Divide both sides by 2:

$$x + 4 = 5 \implies x = 1$$

19 **Answer: C** $\left(-\frac{1}{2}, -\frac{1}{2}\right)$

The intersection of the diagonals of a parallelogram/quadrilateral is the midpoint of either diagonal. Using endpoints $(-5, 3)$ and $(4, -4)$:

$$M = \left(\frac{-5+4}{2}, \frac{3+(-4)}{2}\right) = \left(-\frac{1}{2}, -\frac{1}{2}\right)$$

20 **Answer: A** A median

A median connects a vertex to the midpoint of the opposite side. Both resulting smaller triangles share the same height and have equal bases, so their areas are equal.

21 **Answer: D** 3.5%

Calculate the ratio and convert to percentage:

$$\text{Percentage} = \frac{490}{14000} \times 100\% = 0.035 \times 100\% = 3.5\%$$

22 **Answer: B** $\frac{7}{12}$

Use the Pythagorean identity $\sin^2 x + \cos^2 x = 1$:

$$\sin^2 x = 1 - \cos^2 x = 1 - \frac{5}{12} = \frac{7}{12}$$

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23 **Answer: A** Pairs of same-side interior angles are supplementary.

When parallel lines are cut by a transversal, same-side (consecutive) interior angles add up to $180°$ (supplementary). Alternate interior, alternate exterior, and corresponding angle pairs are congruent.

24 **Answer: D** $A(-4, -3)$

Use the section formula for ratio $1 : 3$ (where $m = 1, n = 3$):

$$x_A = \frac{3(-7) + 1(5)}{1+3} = \frac{-21+5}{4} = \frac{-16}{4} = -4$$

$$y_A = \frac{3(-5) + 1(3)}{1+3} = \frac{-15+3}{4} = \frac{-12}{4} = -3$$

So $A = (-4, -3)$.

25 **Answer: A** 4 boys and 6 girls

Let b be the number of boys and g be the number of girls. $g = 10 - b$.

$$1.75b + 2.25(10 - b) = 20.50$$

$$22.50 - 0.50b = 20.50 \implies 0.50b = 2.00 \implies b = 4$$

So there are 4 boys and 6 girls.

26 **Answer: D** $X > 350,000$

Set up the inequality:

$$38000 + 0.02X > 45000$$

$$0.02X > 7000 \implies X > \frac{7000}{0.02} = 350,000$$

27 **Answer: C** $y = \frac{1}{3}x + 2$

Rewrite $3x + y = 6$ as $y = -3x + 6$. Its slope is $m_1 = -3$.
The perpendicular slope is $m_2 = -\frac{1}{-3} = \frac{1}{3}$.
With y-intercept $c = 2$, the line equation is $y = \frac{1}{3}x + 2$.

28 **Answer: A** $x \geq 2$ and $x \leq 8$

The graph shows solid endpoints at 2 and 8 with the segment in between shaded, representing the intersection $x \geq 2$ and $x \leq 8$ (or $2 \leq x \leq 8$).

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29 **Answer: B** 4 hours

Set the cost equations equal:

$$5h + 50 = 10h + 30$$

$$20 = 5h \implies h = 4 \text{ hours}$$

30 **Answer: D** The set of positive integers

Whole numbers are non-negative integers $\{0, 1, 2, 3, \dots\}$. Nonzero integers exclude 0. Their intersection is $\{1, 2, 3, \dots\}$, which is the set of positive integers.

31 **Answer: C** $66°$

Drawing a parallel auxiliary line through B parallel to lines FAD and EHC splits $\angle ABC$ into two alternate interior angles:

$$m\angle ABC = m\angle FAB + m\angle ECB = 48° + 18° = 66°$$

32 **Answer: A** $C = \frac{E-3}{X}$

Subtract $\frac{3}{X}$ from both sides:

$$C = \frac{E}{X} - \frac{3}{X} = \frac{E-3}{X}$$

33 **Answer: A** 2

Expand the right-hand side:

$$2x(2x + 1) = 4x^2 + 2x$$

Comparing coefficients of x gives $a = 2$.

34 **Answer: C** $90°$

Supplement of $A = 180° - A$.
Complement of $A = 90° - A$.

$$\text{Difference} = (180° - A) - (90° - A) = 90°$$

35 **Answer: C** Graphs A and B are functions.

Step 1: Vertical line test: a scatter plot is a function if no two points share the same x-value.
Step 2: Graphs A and B have exactly one point above each x-value.
Step 3: Graph C has two points with the same x-value, so it is not a function. Answer: Graphs A and B.

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36 **Answer: A** 0.04

To maximize $\frac{x}{y}$, select the largest numerator $x = 0.2$ and the smallest denominator $y = 5$:

$$\text{Max } \frac{x}{y} = \frac{0.2}{5} = 0.04$$

37 **Answer: A** $y = -\frac{3}{4}x + \frac{37}{4}$

Slope $m = \frac{4-7}{7-3} = -\frac{3}{4}$.
Using point-slope with $(3, 7)$:

$$y - 7 = -\frac{3}{4}(x - 3) \implies y = -\frac{3}{4}x + \frac{9}{4} + \frac{28}{4} = -\frac{3}{4}x + \frac{37}{4}$$

38 **Answer: D** 1

"All squares are rectangles" is TRUE (value 1). "All squares are circles" is FALSE (value 0). For an OR logical disjunction, $1 \vee 0 = 1$.

39 **Answer: A** lies inside the triangle

The altitude to the hypotenuse originates from the right-angle vertex and meets the hypotenuse perpendicularly at an interior point, so it always lies completely inside the triangle.

40 **Answer: B** 2,160

Sum of interior angles formula:

$$S = (n - 2) \times 180^\circ = (14 - 2) \times 180^\circ = 12 \times 180^\circ = 2,160^\circ$$

41 **Answer: B** $a^{12}b^{10} - a^2b^6$

Divide each term in the numerator by the denominator a^3b^2:

$$\frac{a^{15}b^{12}}{a^3b^2} - \frac{a^5b^8}{a^3b^2} = a^{12}b^{10} - a^2b^6$$

42 **Answer: C** 24π

Arc length formula: $\frac{60^\circ}{360^\circ} \cdot 2\pi r = \frac{\pi r}{3} = 4\pi \implies r = 12$ cm.
Sector Area $= \frac{60^\circ}{360^\circ} \cdot \pi r^2 = \frac{1}{6} \cdot \pi(144) = 24\pi$ cm^2.

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43 **Answer: A** 3

Using similar triangles $\triangle ADE \sim \triangle ABC$:

$$\frac{DE}{BC} = \frac{AE}{AC} \implies \frac{4}{6} = \frac{AE}{9} \implies AE = 6$$

$$EC = AC - AE = 9 - 6 = 3$$

44 **Answer: D** 25:1

The largest square has side $s_1 = 10$ cm. Moving inwards across 4 gaps of 1 cm on both sides gives the side of the smallest square:

$$s_5 = 10 - 2(4 \times 1) = 2 \text{ cm}$$

Ratio of areas $= 10^2 : 2^2 = 100 : 4 = 25 : 1$.

45 **Answer: A** $-4h^3 - 33h^2 + 31h - 3$

Expand the polynomials:

$$h^2(-4h + 3) + 9h(-4h + 3) - 1(-4h + 3)$$

$$= -4h^3 + 3h^2 - 36h^2 + 27h + 4h - 3 = -4h^3 - 33h^2 + 31h - 3$$

46 **Answer: A** $(x - 20)(x + 3)$

Find two integers with product -60 and sum -17: -20 and $+3$.

$$x^2 - 17x - 60 = (x - 20)(x + 3)$$

47 **Answer: D** $r = \frac{A-p}{pt}$

Subtract p from both sides and then divide by pt:

$$A - p = prt \implies r = \frac{A - p}{pt}$$

48 **Answer: C** q^3

Apply exponent rules:

$$(q^4)^{-3} = q^{-12}$$

$$\frac{q^{-12}}{q^{-15}} = q^{-12-(-15)} = q^3$$

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49 **Answer: C** $20\ \text{cm}^2$

Area of 1 square $= 2 \times 2 = 4\ \text{cm}^2$.
Total area of 5 squares $= 5 \times 4 = 20\ \text{cm}^2$.

50 **Answer: D** $55°$

Since $\overline{XY}$ is tangent to the circle at Y, $HY \perp XY$, so $m\angle HYX = 90°$.
The interior angle $m\angle HXY = 180° - 145° = 35°$.
In right triangle $\triangle HXY$:

$$m\angle XHY = 180° - (90° + 35°) = 55°$$

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